

Let  $p$  be a prime number.

Lemma.  $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \dots + x + 1$

is irreducible.

Proof. observe that  $\Phi_p(x+1) = [(x+1)^p - 1] / (x+1 - 1) = \frac{1}{x} \cdot (x^p + \binom{p}{1}x^{p-1} + \dots + \binom{p}{p-1}x + \binom{p}{0} - 1)$   
 $= x^{p-1} + \binom{p}{1}x^{p-2} + \dots + p,$

so applying the Eisenstein Criterion, we obtain the result.

Lemma. Let  $\zeta_p \stackrel{\text{def}}{=} \exp(2\pi i \cdot \frac{1}{p})$ . Then, the set

$$\{\zeta_p^0, \zeta_p^1, \dots, \zeta_p^{p-1}\} \cong \mathbb{Z}_p.$$

Moreover,  $\mathbb{Q}(\zeta_p)$  is the splitting field of  $x^p - 1$  and also  $\Phi_p$ , with  $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p-1$ .

Proof. It is trivial, since  $p$  is a prime number.  $\square$

Lemma.  $[\mathbb{Q}(\zeta_p + \zeta_p^{-1}) : \mathbb{Q}] = \frac{p-1}{2}$ .

Proof. Since  $\zeta_p^{-1} = \zeta_p^{p-1} = \exp(2\pi i \cdot \frac{p-1}{p})$ ,  $\zeta_p + \zeta_p^{-1} = 2 \cdot \cos(2\pi \cdot \frac{1}{p}) \in \mathbb{R}$

Then,  $x^2 - (\zeta_p + \zeta_p^{-1})x + 1 \in \mathbb{Q}(\zeta_p + \zeta_p^{-1})[x]$  and it has roots as  $\zeta_p$  and  $\zeta_p^{-1}$ .

For odd prime  $p$  ( $p \geq 3$ ),  $\zeta_p \notin \mathbb{Q}(\zeta_p + \zeta_p^{-1})$ , so the polynomial above is irreducible.

Therefore,  $[\mathbb{Q}(\zeta_p) : \mathbb{Q}(\zeta_p + \zeta_p^{-1})] = 2$ . Now, the tower law gives the result.

(If  $p = 2$ , then  $\zeta_2 = -1$ , so proved.)

Thm. Splitting field of  $x^{p-2}$  over  $\mathbb{Q}$  is  $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$ , and  $[\mathbb{Q}(\sqrt[p]{2}, \zeta_p) : \mathbb{Q}] = p(p-1)$ .

Proof. Since for each  $0 \leq i \leq p-1$ ,  $(\sqrt[p]{2} \cdot \zeta_p^i)^p = 2 \cdot \zeta_p^{pi} = 2 \cdot 1 = 2$ ,

so the splitting field of  $x^{p-2}$  is  $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$ . To evaluate the degree, observe that

$$\mathbb{Q}(\sqrt[p]{2}, \zeta_p) = \mathbb{Q}(\sqrt[p]{2})\mathbb{Q}(\zeta_p) \quad \text{and}$$

$$[\mathbb{Q}(\sqrt[p]{2}) : \mathbb{Q}] = p \quad \text{and} \quad [\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p-1, \quad \text{so}$$

$$[\mathbb{Q}(\sqrt[p]{2}, \zeta_p) : \mathbb{Q}] = p(p-1).$$

Example.  $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$

The  $K$  is a splitting field for  $(x^2-2) \cdot (x^2-3)$



But, the splitting field of  $x^3-2$  is not just  $\mathbb{Q}(\sqrt[3]{2})$ ,

because there are three roots:

$$\sqrt[3]{2}, \quad \sqrt[3]{2} \cdot \left( \frac{-1 + i\sqrt{3}}{2} \right), \quad \sqrt[3]{2} \cdot \left( \frac{-1 - i\sqrt{3}}{2} \right)$$

Let  $K$  be the splitting field of  $x^3-2$ . Then is,

$$\sqrt[3]{2} \in K \quad \text{and} \quad \sqrt[3]{2} \cdot \left( \frac{-1 + i\sqrt{3}}{2} \right) \in K,$$

so  $\sqrt{-3} \in K$ . It follows that  $K = \mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$ .

Since  $\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3}) = \mathbb{Q}(\sqrt[3]{2}) \mathbb{Q}(\sqrt{-3})$  and

$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$  and  $[\mathbb{Q}(\sqrt{-3}) : \mathbb{Q}] = 2$ , these are relatively prime,  
thus the degree of composite field is 6.

Find the Galois group of  $x^5 - 3 \in \mathbb{Q}(\xi)[x]$  over  $\mathbb{Q}(\xi)$ , where  $\xi$  is a primitive fifth root of unity.

Proof. Let  $K$  be the splitting field of  $x^5 - 3$ .

Since all roots of  $x^5 - 3$  are

$$\sqrt[5]{3} \cdot \xi^k \quad (k=0,1,2,3,4) \quad \text{where} \quad \xi = \exp\left(2\pi i \cdot \frac{1}{5}\right) = e^{\frac{2\pi i}{5}},$$

Therefore  $K = \mathbb{Q}(\sqrt[5]{3}, \xi)$ . Now,

$$K = \mathbb{Q}(\sqrt[5]{3}, \xi)$$

$$\begin{array}{cc} 4 & \backslash & 5 \\ \mathbb{Q}(\sqrt[5]{3}) & & \mathbb{Q}(\xi) \end{array}$$

$\because x^5 - 3$  is irreducible over  $\mathbb{Q}$  (by Eisenstein with  $p=3$ ) and  $x^4 + x^3 + x^2 + x + 1$  has root  $\xi$  and is irreducible over  $\mathbb{Q}$ .

Since  $[\mathbb{Q}(\sqrt[5]{3}, \xi) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[5]{3}) : \mathbb{Q}] \cdot [\mathbb{Q}(\xi) : \mathbb{Q}]$  and both degrees over  $\mathbb{Q}$  are relatively prime,  $[K : \mathbb{Q}] = 20$ . And, since  $[K : \mathbb{Q}(\xi)] = 5$ , prime,

$$\text{Gal}(K/\mathbb{Q}(\xi)) \cong \mathbb{Z}_5.$$